RA01 CYBERPUNK VR

Controls Info Sheet

Every button, stick and body movement, and what reaches the game

Written for game build 2.31 - 12 October 2026 - source: Core\controls_mapping\ControllerMapping.cs

1. How the controls work (one paragraph)

The game thinks an Xbox pad is plugged in. Each time it asks Windows for the pad (`XInputGetState`, slot 0), **XInputHook** answers instead with a pad built by **ControllerMapping.Build()** from your motion controllers (**Controllers.cs**, read through OpenXR every headset frame). Before each build, the Actions features run first. They may add a press (a melee strike taps RT), hold one (shouldering a gun holds LT), or keep one back (gesture mode keeps your triggers). They always add on top of your own controls and never replace them. Any other pad slot passes straight through to Windows.

> Builder picture: a control-panel re-label, not a rewiring. The game still sees an ordinary Xbox panel. RA01 decides which of your real switches is wired to each label, and a few automatic hands press labels for you when your body does the move.

2. Default layout (right-handed)

Names follow Meta Touch controllers: left hand X / Y, right hand A / B.

Your control	Game pad button	What it does in play
Left stick	Left stick	Move.
Left stick click	L3	The game's L3 action (sprint in the standard layout).
Right stick left / right	Right stick X	Turn. "Body follows head" adds your head turn on top (section 5).
Right stick up / down	*(not sent)*	Off by default: your head does the looking. Turn on "Stick look up/down" for seated play. Game menus always get it.
Right stick quick click (under 0.3 s, left stick untouched)	R3, short press	Crouch (the game's crouch must be set to Toggle).
Right stick held in + left stick	D-pad up / down / left / right	The D-pad modifier, since Touch has no D-pad. While held this way, the click is never sent.
Left trigger	LT	Aim / block.
Right trigger	RT	Fire / attack. Hold RT to charge (power attack, Monowire).
Left grip (past halfway)	LB	The game's LB: a short press is the quick melee with a gun, a long press is the scanner. Held back while your off hand reaches for or holds a weapon, or carries a magazine.
Right grip (past halfway)	RB	The game's RB: hold for a grenade / throwable.
X	X	Interact / reload. With Physical reloading on and a gun drawn, X also drops the magazine. Horn while driving.

Your control	Game pad button	What it does in play
Y	Y	Weapon swap. Hold for the weapon wheel. Never swallowed.
A	A	Jump / confirm.
B	B	Back / cancel in menus, and the game's B action in play. With Gesture mode on, during play B reaches the game only as a quick tap (under 0.4 s), so the 5 s hold is the mod's.
Left menu button	Start	Game menu (inventory, character, map, journal).

3. Opening menus and the settings window

Do this	Result
Hold X and LT (X first, or both together), then push the **right stick UP**	Game menu: inventory, character, crafting, map, journal (Start).
... right stick **DOWN**	Pause menu (Back / View).
... right stick **LEFT**	Opens or closes the **F10 settings window**.
F10 on the keyboard	Opens or closes the settings window (in VR, or on the monitor with no headset).
In the window: **point the right controller** like a laser	Moves the pointer. The **right trigger clicks**. Both sticks scroll a page at a time.

On every controller the menu chord works the same way: X + LT on the X side, then the turning stick (in left-hand mode: right A + right trigger, then the left stick). While the chord is held the game gets **no LT**, so V doesn't aim. LT pressed first and then X is aiming plus reloading, not the chord. While the F10 window is open, the sticks and the clicking trigger never reach the game, so V doesn't walk, turn or fire.

4. Physical controls (your body)

Feature (F10 tab)	Do this	What reaches the game
Physical aiming (Weapons)	Point the gun itself.	Shots leave the muzzle along the barrel while you pull RT (AimProbe / ShotProbe).
Shoulder to aim (Weapons)	Raise the gun to your cheek, pointing where you look.	LT held for you (no zoom, with No aim zoom on).
Right stick D-pad while aiming (Weapons)	Hold LT, push the right stick up / down.	D-pad up / down. Left / right still turn.
Two-handed grip (Weapons)	Reach your off hand to the gun and squeeze its grip.	The gun turns along the line between your hands. LB / RB are held back.
Virtual stock (Weapons)	Bring the back of the gun to your shoulder.	Aim steadied along the shoulder-to-hand line.

Feature (F10 tab)	Do this	What reaches the game
Physical reloading (Weapons, off by default)	**Pull:** off hand on the magazine with its grip open, push and **hold the off-hand trigger** and pull down. It's in your hand until you let go of the trigger, then it drops. Or press X to drop it. **New one:** grip **or** trigger at your off-hand hip (the glowing ball), carry it to the small ball under the gun.	The game's own refill. While you're reloading, the off-hand grip and trigger send nothing to the game (no aiming, no LB). Gripping the gun with the off hand = the two-handed grip, and its trigger aims as normal.
Physical melee (Weapons, off by default)	Swing fast to strike. Hold a guard pose in front of your face to block.	RT tapped / LT held. A fast swing with a gun taps LB (quick melee).
Physical power attacks (Weapons)	Pull a hand back behind your shoulder, hold 0.1 s, swing. Gorilla Arms: both hands above your head, then slam down.	RT held while it loads, released on the swing (the strong attack).
Physical grenade throw (Weapons)	Hold the grip, swing, let go of the grip as your hand comes forward.	The grenade lands where a real throw would (ThrowLever order).
Physical knife throw (Weapons)	Hold to aim, then swing the knife hand forward hard.	The throw is pressed for you, along your swing.
Physical crouch (VR Controls)	Squat for real (30 cm below your recentred height by default).	R3 tapped to crouch, tapped again to stand.
Physical swimming (VR Controls)	Pull your arms back to swim. Sweep a hand out to turn the other way. Look down to dive.	Left stick forward plus a right-stick turn. Your thumb always wins.
Physical ladder (VR Controls)	Squeeze a grip on a rung and pull down to climb up.	Left stick up / down. Grips are not LB / RB on a ladder.
Physical steering (Vehicle)	Hand on the wheel rim or handlebar, squeeze the grip, turn. Palm on the hub = horn.	Left stick X. X held for the horn. Grips are not LB / RB while seated.
Gesture mode (VR Controls)	Hold B for 5 s to switch on or off. Then: grip = point, trigger = thumbs up, both = middle finger, nothing = open hand.	Nothing. Triggers and grips pose V's fingers only. Drawing a weapon switches it off.

5. Context changes (the same button, a different job)

Situation	What changes
Body follows head (on, dead zone 40 degrees)	Look freely within 40 degrees each side of V's body. Past it, a right-stick push is added until V's body lines up under your head again (within 5 degrees). The world stays still. Your real right stick always wins.
Keep view level (on)	If a cutscene leaves the game camera tilted more than 1.5 degrees for 0.75 s, a right-stick up / down push levels it. It is never active in menus or vehicles, or when the camera isn't V's own.
Braindance / taken-over camera	The right stick looks up / down as well. Body follow and the camera height shift pause.
Driving with a gun out	RT and RB swap. Your gun hand's trigger is RB and its grip squeeze is RT (the throttle, as deep as you squeeze).
In vehicles (left-hand mode)	Left trigger = throttle, right trigger = brake / reverse.

Situation	What changes
Swimming / ladder / steering	The mod drives the left stick, but any real thumb movement past 10% overrides it at once. If the feature stops talking for 150 ms, everything goes back to normal by itself.
F10 window open	All four stick axes are zeroed for the game, and the right trigger (left in left-hand mode) is the window's click.
Game menus	Triggers and grips keep their normal places. The right stick's up / down is always sent.

6. Left-hand mode (VR Controls tab)

One switch mirrors everything. **Read before switching:** it changes how to open the settings window too.

Action	Right-handed (default)	Left-hand mode
Fire / attack (RT)	Right trigger	**Left trigger**
Aim / block (LT)	Left trigger	**Right trigger**
Move / turn	Left stick / right stick	**Right stick / left stick**
Crouch (R3 quick click)	Right stick click	**Left stick click**
LB / RB	Left grip / right grip	**Right grip / left grip** (in play only)
Reload (X) / jump (A)	Left X / right A	**Right A / left X** (face buttons swap sides)
Gesture mode hold (B)	Right B	**Left Y**
Menu chord: game menu	Hold X + LT, right stick UP	**Hold right A + right trigger, left stick UP**
Menu chord: pause menu	Hold X + LT, right stick DOWN	**Hold right A + right trigger, left stick DOWN**
Menu chord: settings window	Hold X + LT, right stick LEFT	**Hold right A + right trigger, left stick LEFT**
F10 pointer hand	Right	**Left**
Weapon hand	Right	**Left.** The mod moves each weapon across every frame, because the game only animates right-handed.

7. Other headsets' controllers (M50)

Section 2 uses Meta Touch names. RA01 also gives the runtime its own button list for each controller below (`Core\controls_mapping\Controllers.cs`, M50). Each list stands alone: if a runtime refuses one, only that one is skipped. The non-Touch layouts are **not yet tested on real hardware**.

> **Not supported: HTC Vive wands** (the original Vive / Vive Pro stick-shaped controllers with round trackpads). They have no thumbsticks and too few buttons for this game. A Vive headset works with Index (or other supported) controllers. The Vive XR Elite's controllers are Touch-style.

7.1 Right-handed (default)

Game button	Meta Touch	Valve Index	Windows MR	HP Reverb G2	Steam Frame
Move / turn	Sticks	Sticks	Sticks	Sticks	Sticks
L3 / R3	Stick clicks	Stick clicks	Stick clicks	Stick clicks	Stick clicks
D-pad	Right stick click + left stick	same	same	same	**Left D-pad** (the stick-click trick is off; right-stick click = crouch only)
LT / RT	Triggers	Triggers	Triggers	Triggers	Triggers
LB / RB	Grips	Grips	Grips (on / off)	Grips	Grips, **and** bumpers (bumpers only press LB / RB)
X	Left X	Left A	Left pad click	Left X	**Right X**
Y	Left Y	Left B	none	Left Y	**Right Y**
A	Right A	Right A	Right pad click	Right A	Right A
B	Right B	Right B	Right menu	Right B	Right B
Start (game menu)	Left menu	Left trackpad press	Left menu	Left menu	**Right Menu**
Back (pause menu)	chord	chord	chord	chord	**Left View, quick tap**
F10 window	X, LT, right stick left	same	same	same	**Hold left View 1 s**

7.2 Left-hand mode

Sticks, triggers and grips mirror on every controller, as in section 6. What changes per controller:

Controller	Left-hand mode
Meta Touch, HP Reverb G2	Face buttons swap sides (right A = X, right B = Y, left X = A, left Y = B), as in section 6. Start stays on the left menu.
Valve Index	A / B swap sides the same way. **Start moves to the right trackpad press.**
Windows MR	Pad clicks swap (right pad = X, left pad = A). **The menu buttons swap: left menu = B, right menu = Start.** No Y.
Steam Frame	**A / B / X / Y and the D-pad trade places by position:** your left D-pad becomes A (down), B (right), X (left) and Y (up), and the right A / B / X / Y become the game's D-pad. So you can jump with your left thumb while your right thumb moves. Bumpers mirror like the grips. Menu (Start) stays right; View (Back / F10) stays left.

7.3 Notes

- **Menu chord** (X + LT, then the turning stick: up = game menu, down = pause, left = settings window) works on Touch, Index, Windows MR and Reverb G2, right- and left-handed, because X is always on the opposite thumb from the stick you push. The **Steam Frame** doesn't need it: Menu = game menu, View tap = pause, View held 1 s = settings window.
- **Steam Frame:** its own list is used when SteamVR offers the `XR_VALVE_frame_controller_interaction` extension. Otherwise SteamVR maps the Touch list onto it.
- **HP Reverb G2:** its own list is used when the runtime offers `XR_EXT_hp_mixed_reality_controller`. Otherwise it uses the Windows MR layout.
- **Which layout is in use** is asked of the runtime once a second. The Core VR log says `controller layout in use - Frame / Index / Wmr / Other`. Touch and G2 count as "Other", which is the default layout.
- **Hand angle:** each controller holds its "hand point" at a slightly different angle. On a new controller, run "Calibrate arms (T-pose)" first. If a gun still looks tilted, report it.
- **Log check:** each list the runtime accepted logs `RA01 controllers: bindings suggested for /interaction_profiles/...`.

8. Keyboard and mouse

Key	What it does
Home	Recentre now: your head's spot, facing and height become the centre.
End	Head steering on / off (quick physical toggle).
F10	Settings window open / close.
Left mouse button	Clicks inside the F10 window (the mouse moves its pointer).
F11	Dev: one-off read-only skeleton heap hunt (BoneFinder). Retired tool.
F9	**Not bound by RA01.** It is Cyberpunk's quickload. See the note below.

> **Found while writing this:** the "Game menu view" row's help text in SettingsMenu.cs still says "The F9 key switches between Menu and Auto", but no code reads F9 any more (it was removed in M5a-12, and Keys.cs says F9 is the game's quickload). The text is stale. A one-line text fix if you want it. **Not changed.**

9. Recentre and calibration

- **Recentre:** Home, the F10 window's Recentre button, or the headset's own recentre (Meta button hold). XrSession treats a moved floor space as a recentre.
- **Standing height** for Physical crouch is taken at each recentre.
- **Calibrate arms (T-pose)** and **Put camera at V's eyes** (VRIK tab): press, then hold the pose or stand still as the status row says. Both are saved.